YIFEI YUAN

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Ph.D. candidate in embodied AI, training physics-based musculoskeletal human agents and co-adaptive wearable-robot controllers with RL, from GPU-parallel simulation to sim-to-real deployment validated with human subjects. First-author papers at IROS, BioRob, and ICNR (all oral), with three more under review; reviewer for ICRA, IROS, and RA-L. Seeking a Summer 2027 research internship.

EDUCATION

New Jersey Institute of Technology & Rutgers University — Newark, NJ
Ph.D. Candidate, Biomedical Engineering (Joint Program) — *Sept 2024 – Expected 2028*
- Advisor: Prof. Xianlian A. Zhou · Co-advisor: Dr. Ghaith Androwis (Kessler Foundation) · Proposal defended May 2026.

Washington University in St. Louis — St. Louis, MO
M.S. in Information Systems (GPA 3.93/4.00); M.Eng. in Engineering Management — *Sept 2022 – May 2024*

Southwest Petroleum University — Chengdu, China
B.Eng. in Telecommunications Engineering (GPA 3.58/4.00) — *Sept 2018 – June 2022*

RESEARCH EXPERIENCE

BioDynamics Lab, New Jersey Institute of Technology — Newark, NJ
Graduate Research Assistant (Advisor: Prof. Xianlian A. Zhou) — *Aug 2024 – Present*

UniExo – unified multi-skill musculoskeletal locomotion & co-adaptive exoskeleton control — *under review*
- **Scalable training:** built a GPU-parallel RL stack (MuJoCo Warp + mjlab + rsl_rl) that makes high-dimensional control of an 80-muscle musculoskeletal model tractable: 8,192 parallel envs on one GPU at 3.7×10^4 env steps/s.
- **Unified human policy:** distilled walk, turn, run, and backward experts into one latent-conditioned muscle-actuated policy via DAgger + RL fine-tuning, switching skills without a mode signal at 94.7% tracking success on unseen clips.
- **Unified exo controller:** replaced activity-specific exoskeleton control with a single hip-exo controller co-adapted with the human policy via multi-agent RL, assisting all four skills and their transitions without mode switching.
- **Human-subject validation:** designed the IRB-approved protocol and deployed the controller with no per-user retuning, generalizing across 6 participants, 4 treadmill speeds, and a continuous walk–turn–run–backward route.
- **Embedded electronics:** designed the exoskeleton's Raspberry Pi 5 real-time controller, CAN-bus communication with MyActuator motors, multi-IMU real-time sensing, and the power distribution system.

SMAT – staged multi-agent training for co-adaptive exoskeleton control — *IROS 2026 (Oral); IEEE T-ASE under review*
- **Co-adaptive MARL framework:** designed a staged CTDE multi-agent RL framework (human actor, exo actor, shared critic) with a 4-stage PPO curriculum that co-trains a 26-muscle human model and a hip-exo controller in simulation.
- **Zero-shot sim-to-real:** transferred the sim-trained controller to a physical hip exoskeleton using IMU-only sensing and no hardware-side parameter tuning, establishing a practical sim-to-real pipeline for co-adaptive assistance.
- **Metabolic benefit:** reduced metabolic cost by 19.7% vs. a passive device in human-subject trials, validating simulation-trained co-adaptation as a practical path that avoids lengthy hand-tuning or human-in-the-loop optimization.

Kessler Foundation — West Orange, NJ
Research Assistant (Supervisor: Dr. Ghaith Androwis) — *Mar 2025 – Present*
- **Exoskeleton electronics:** contributed to the design of an 8-DoF lower-limb rehabilitation exoskeleton; led its overall electronics and control system and integrated the force and IMU sensors.
- **Full-body sim-to-real:** trained a walking controller with a 304-muscle full-body musculoskeletal model in DART and deployed it from onboard sensing only via a PyTorch→ONNX→MATLAB real-time pipeline (ICNR 2026, Best Oral Award).
- **Sensor fusion:** built an IMU + motion-capture pipeline for real-time monitoring of human–exoskeleton interaction.

Franken Lab, Department of Neuroscience, Washington University School of Medicine — St. Louis, MO
Research Assistant (PI: Dr. Tom Franken) — *Oct 2023 – Aug 2024*
- Collected multi-channel neural electrophysiology data from non-human primates with multi-electrode probes.
- Built spike-sorting, clustering, and data quality-control pipelines in Python and MATLAB.

INDUSTRY EXPERIENCE

Erebus Technologies – Electrical Engineer & Technical Advisor Intern, St. Louis, MO — *May 2023 – Sept 2023*
- Clustered mobile-device displacement data to reconstruct movement paths; designed circuitry and drafted patent sections.

iQIYI – Software Engineer Intern, China — *Mar 2022 – June 2022*
- Ran pre-release QA on movie and TV video-bitstream uploads, ensuring every stream cleared ingestion before release.
- Refined recognition scripts for the team's OCR platform to improve text-recognition accuracy.

PROJECTS

LLM-Driven Robot Navigation Agent – ROS 2, Gazebo, Llama/Gemini — *Jan 2025 – May 2025*
- Built an agent that interprets natural-language commands, reasons over robot state, and plans navigation in simulation.

Human Motion Prediction from Wearable IMUs — *Dec 2020 – Apr 2022*
- Collected data from 17 body-worn IMUs and benchmarked RNN, LSTM, 1D-CNN, and MLP models for sequential motion forecasting; LSTM achieved the highest accuracy.

Publications

[1] **Y. Yuan**, J. Wolf, G. Androwis, X. Zhou. "Multi-Agent Reinforcement Learning of Co-Adaptive Hip Exoskeleton Control (MARL-Exo) for Metabolic Energy Reduction." *Under review at IEEE*, 2026. arXiv:2610.37441

[2] **Y. Yuan**, J. Wolf, G. Androwis, X. Zhou. "UniExo: Unified Multi-Skill Policies for Musculoskeletal Locomotion and Co-Adaptive Exoskeleton Control." *Under review*, 2026. arXiv:2609.19690

[3] **Y. Yuan**, J. Wolf, G. Androwis, X. Zhou. "Staged Multi-Agent Training (SMAT) for Hip Exoskeletons: Metabolic and Biomechanical Validation of a Simulation-Trained Co-Adaptive Controller." *Under review at IEEE Transactions on Automation Science and Engineering (T-ASE)*, 2026. arXiv:2608.00715

[4] **Y. Yuan**, G. Androwis, X. Zhou. "SMAT: Staged Multi-Agent Training for Co-Adaptive Exoskeleton Control." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. **Oral.** arXiv:2603.07618

[5] **Y. Yuan**, G. Androwis, X. Zhou. "Gait Asymmetry from Unilateral Weakness and Improvement with Ankle Assistance: A Reinforcement Learning Based Simulation Study." *IEEE RAS/EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob)*, 2026. **Oral.** IEEE Xplore · arXiv:2602.18862

[6] **Y. Yuan**, E. H. Nunez, E. S. Suviseshamuthu, S. Adamovich, G. Androwis, X. Zhou. "Sim-to-Real Deployment of a Deep Reinforcement Learning Controller for a Lower-Limb Rehabilitation Exoskeleton." *International Conference on NeuroRehabilitation (ICNR)*, 2026. **Best Oral Presentation Award.**

[7] Z. You, **Y. Yuan**, M. Tohfafarosh, X. Zhou. "Fast Personalization of Real-Time Human Activity Recognition via Transfer Learning." *IEEE BioRob*, 2026. **Oral.** IEEE Xplore

[8] **Y. Yuan**, X. Zhou. "Ankle Assistance Restores Gait Symmetry under Unilateral Muscle Weakness: A Reinforcement Learning Based Simulation Study." *IISE Annual Conference*, 2026. **Oral.**

[9] **Y. Yuan**, X. Zhou. "Unilateral Muscle Weakness Induces Progressive Gait Asymmetry: A Reinforcement Learning Based Simulation Study." *Northeast Bioengineering Conference (NEBEC)*, 2026. Poster.

Talks

- "SMAT: Staged Multi-Agent Training for Co-Adaptive Exoskeleton Control." IEEE/RSJ IROS, Pittsburgh, PA, Sept 2026.
- "Gait Asymmetry from Unilateral Weakness and Improvement with Ankle Assistance." IEEE BioRob, Edmonton, Aug 2026.
- "Fast Personalization of Real-Time Human Activity Recognition via Transfer Learning." IEEE BioRob, Edmonton, Aug 2026.
- "Ankle Assistance Restores Gait Symmetry under Unilateral Muscle Weakness." IISE Annual Conference, 2026.

Professional Service

- Reviewer: IEEE International Conference on Robotics and Automation (ICRA); IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE Robotics and Automation Letters (RA-L).

Technical Skills

Learning	PPO, SAC, multi-agent RL (CTDE), imitation learning (DAgger, AMP), policy distillation, domain randomization
Simulation	MuJoCo, MuJoCo Warp, mjlab, DART, Isaac Lab, Isaac Gym, Genesis, MyoSuite, ROS 2, Gazebo, MJCF modeling
Hardware	Raspberry Pi control, CAN bus, brushless actuators, IMUs, force/torque sensors, motion capture
Languages	Python, C++, MATLAB, Shell; PyTorch, ONNX, rsl_rl